

A Project Report On

Marketa

Submitted in partial fulfillment of the requirement for the
award of the degree

Master of Computer Applications
(MCA)

Academic Year 2026 – 27

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Marwadi
University
Marwadi Chandarana Group



Faculty of Computer Applications (FoCA)

Certificate

This is to certify that the project work entitled

Marketa

submitted in partial fulfillment of the requirement for

the award of the degree of

Master of Computer Applications (MCA)

of the

Marwadi University

is a result of the bonafide work carried out by

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Dean

DECLARATION

We hereby declare that this project work entitled Markets is a record done by us.

We also declare that the matter embodied in this project is genuine work done by us and has not been submitted whether to this University or to any other University for the fulfillment of the requirement of any course of study.

Place:

Date:

Nevil Aghera 92500584012 Signature:_____

Jenil Kothiya 92500584026 Signature:_____

Avesh Kadari 92500584027 Signature:_____

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Avesh Kadari	92500584027	Signature:_____

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1. Introduction to Project Definition

MARKETA is a full-stack, enterprise-grade web application engineered using the **MERN Stack (MongoDB, Express.js, React.js, Node.js)** that digitizes, streamlines, and automates the complete operational workflow of modern supermarkets and retail grocery stores.

In traditional retail environments, store managers and cashiers heavily rely on manual ledger books, disconnected spreadsheet files, physical receipts, and fragmented desktop software to manage daily business activities. These conventional methods present major operational bottlenecks:

- **Manual Inventory Errors:** Manual stock counting is time-consuming, highly susceptible to human miscalculations, and often leads to unrecorded inventory shrinkages or stockouts during peak business hours.
- **Complex Supplier & Purchase Management:** Tracking vendor purchase orders, payment dues, and stock receipts across multiple suppliers leads to miscommunication and delayed inventory restocking.
- **Inefficient Checkout & Billing Queue:** Manual pricing lookups and slow invoice generation result in long customer waiting times at billing counters.
- **Lack of Real-Time Financial Visibility:** Calculating net profits, gross margins, daily revenue, and tax liabilities manually at the end of each day or month is cumbersome and prone to financial discrepancies.

MARKETA solves these challenges by providing a unified, cloud-ready digital ecosystem where all supermarket operations—ranging from product cataloging, supplier management, and POS billing to automated inventory updates, low-stock alert notifications, and multi-dimensional financial analytics—are managed seamlessly through an intuitive single-window dashboard.

By implementing strict Model-View-Controller (MVC) architectural separation on the backend and component-driven state management on the frontend, the system provides high throughput, real-time data synchronization, multi-role security (Admin and Manager access controls with pending approval gates), and zero-licensing cost deployment.

2. PREAMBLE

2.1 Synopsis :

MARKETA is a cloud-native Supermarket Management System designed and implemented using the MERN Stack. The application is built to transition brick-and-mortar retail enterprises into digitally managed, highly efficient smart supermarkets by eliminating manual paper-based processes and fragmented software workflows.

The system empowers supermarket managers, store owners, and cashiers to manage every facet of retail operations from a centralized, web-responsive dashboard. Key operational capabilities include:

1. **Catalog & Category Management:** Hierarchical product organization with barcode mapping, unit classification, purchasing cost, and selling price configuration.
2. **Point of Sale (POS) & Billing:** High-speed cashier billing terminal supporting item scanning, custom discount calculation, tax processing, receipt printing, and real-time inventory deduction.
3. **Supplier & Purchase Order Tracking:** Complete procurement lifecycle tracking from supplier contact management to purchase invoice recording and stock intake auditing.
4. **Automated Stock Monitoring & Alerts:** Real-time stock level monitoring that calculates low-stock thresholds (isLowStock) and out-of-stock states (isOutOfStock), automatically triggering high-priority system alerts and notifications.
5. **Customer Profiling & Debt Tracking:** Storing customer purchase history, total expenditure, and contact records for personalized customer relationship management.
6. **Financial Reports & Business Intelligence:** Interactive graphical analytics powered by Recharts displaying sales trends, revenue performance, profit margins, and inventory asset valuation.
7. **Secure Multi-Role Access Control:** Role-based security enforcing strict Admin and Manager permission boundaries, including an automated Admin approval gate for newly registered store managers.

The application follows a modular, decoupled architecture where each sub-module performs a specialized business function while communicating through secure RESTful API endpoints. **MongoDB** provides a flexible, scalable document data store; **Node.js** and **Express.js** execute asynchronous server-side business logic; and **React.js** delivers an interactive, fluid single-page user interface (SPA).

2.2 Problem Statement :

Small to medium-sized supermarkets face severe operational vulnerabilities due to outdated management workflows. The primary problems addressed by MARKETA include:

1. **Inventory Discrepancy & Stockouts:** Lack of real-time stock tracking causes unexpected stockouts of high-demand items, resulting in lost revenue and customer dissatisfaction.
2. **Inaccurate Financial Accounting:** Manual record-keeping leads to calculation errors when computing gross sales, supplier payables, net profit margins, and discounts.
3. **Security Risks & Unauthorized Access:** Traditional desktop systems lack granular audit trails and approval gates, exposing store data to unauthorized edits or unapproved staff actions.
4. **Slow Checkout Processing:** Manual product searching during billing increases customer wait times and checkout counter congestion.
5. **Lack of Decision-Making Analytics:** Store owners lack visual data insights regarding top-selling products, slow-moving items, and monthly sales trends.

2.3 Objectives of the System :

The overarching objective of the MARKETTA Supermarket Management System is to deliver a secure, scalable, and cost-effective digital web platform that automates retail management. Specific technical and business objectives include:

1. **Automate Inventory Lifecycle:** Automatically deduct product stock levels immediately upon POS sale completion and update stock levels when new supplier purchases are recorded.
2. **Implement Real-Time Alerting:** Trigger automated notifications whenever an item's stock drops below its predefined minimum threshold (minStock), enabling timely replenishment.
3. **Accelerate POS Billing & Invoicing:** Provide cashiers with a rapid, intuitive billing interface with discount calculation, tax handling, and instant thermal receipt printing (react-to-print).
4. **Ensure High Security & Access Control:** Enforce JWT-based stateless authentication, password encryption (bcrypt.js), role segregation (Admin vs. Manager), and an Admin approval workflow for new registrations.
5. **Deliver Business Intelligence Analytics:** Present store owners with interactive charts and reports detailing daily sales, top products, customer expenditure, and category revenue distribution.
6. **Eliminate Software Licensing Costs:** Provide an open, self-hostable web architecture that frees small business owners from high recurring SaaS software subscriptions.

2.4 Scope of the System :

A. In-Scope Modules & Features

1. **User Authentication & Admin Approval Gate:** Registration, login, profile fetch, JWT token management, and Admin approval panel for pending managers.
2. **Product Catalog & Stock Management:** Product creation, image upload via Multer, price setup, barcode assignment, category linking, and stock history tracking.
3. **Category & Supplier Management:** Full CRUD operations for product categories and vendor supplier records.
4. **Point of Sale (POS) & Invoice Generation:** Cart management, price auto-calculation, custom discounts, invoice auto-numbering (INV-YYYYMMDD-XXXX), and printing.
5. **Customer Management:** Recording customer contact info, tracking total visits, and calculating total lifetime spending.
6. **Automated System Notifications:** Real-time generation of low_stock and out_of_stock notification logs.
7. **Financial Reports & Dashboard Analytics:** Aggregate revenue calculation, profit metrics, and visual Recharts line/bar graphs.

B. Out-of-Scope (Future Boundaries)

- Direct hardware-level WebUSB barcode scanner driver compilation (currently uses plug-and-play keyboard emulation).
- Multi-store chain inter-branch inventory transfers (currently scoped to single supermarket enterprise).
- Direct integrated online payment gateway webhooks (currently supports cash/card POS billing modes).

2.5 Module description :

1. Authentication Module:

Provides secure login and registration for users. It verifies user credentials and ensures that only authorized managers and administrators can access the system.

2. Dashboard Module:

Displays a real-time overview of the supermarket's operations. It summarizes sales, purchases, inventory, profit, and stock alerts for quick decision-making.

3. Product Management Module:

Manages all product-related information, including adding, updating, deleting, and searching products. It helps maintain accurate product records and stock details.

4. Category Management Module:

Organizes products into different categories for better management. It simplifies product classification and improves product searching.

5. Supplier Management Module:

Stores and manages supplier information used for purchasing products. It helps maintain supplier records and simplifies purchase transactions.

6. Customer Management Module:

Maintains customer details and purchase history. It helps the supermarket provide better customer service and track buying patterns.

7. Inventory Management Module:

Monitors and controls product stock levels. It records stock movements and ensures inventory remains accurate through stock adjustments.

8. Purchase Management Module:

Records all purchases made from suppliers and automatically updates inventory. It maintains purchase invoices and purchase history for future reference.

9. Sales Management Module:

Handles customer sales transactions by generating invoices and reducing stock automatically. It ensures accurate billing and maintains complete sales records.

10. Invoice Management Module

Generates and manages purchase and sales invoices with unique invoice numbers. It provides printable invoices and stores invoice history.

11. Stock Alert Module

Continuously monitors inventory levels and identifies products that are low or out of stock. It notifies the manager to reorder products on time.

12. Financial Management Module

Tracks the supermarket's income, expenses, and overall profit. It provides financial summaries that help managers monitor business performance.

13. Reports Module

Generates detailed reports for sales, purchases, inventory, and customers. These reports help analyze business performance and support decision-making.

14. Shop Settings Module

Allows the manager to configure supermarket information such as shop name, GST number, logo, address, and invoice settings. It personalizes the application for the business.

3. REVIEW OF LITERATURE

Study & Analysis of Existing Systems :

A comparative study of current supermarket inventory and POS management solutions was conducted:

1. **Legacy Desktop ERP Systems (e.g., Marg ERP):** Standalone desktop software. High license cost, single-terminal restriction, complex UI.
2. **Cloud SaaS Solutions (e.g., Square POS / Odoo):** Proprietary SaaS platforms. High recurring monthly fee per terminal, closed architecture.
3. **MARKETA System (Proposed MERN Stack):** Open, web-responsive, self-hostable, zero licensing cost, role-gated with real-time stock sync.

Comparative Analysis Matrix :

Comparison Criteria	Legacy Desktop (Marg ERP)	SaaS Cloud POS (Square/Odoo)	MARKETA System (Proposed MERN)
Architecture	Standalone Desktop (C++/VB)	Vendor Cloud SaaS	Full-Stack MERN Web SPA
Cost Model	High One-Time License Fee	Monthly Recurring Subscription	Free & Self-Hostable (Zero License Fee)
Multi-Device Support	Restricted to Single PC	Web / Mobile (Vendor Locked)	Any Browser (Desktop, Tablet, Mobile)
User Role Access	Basic Admin / User Accounts	Fixed Tier Roles	Granular Admin & Manager with Approval Gate
Stock Auto-Deduction	Manual / Batch Processing	Real-Time Sync	Real-Time Auto-Deduction on POS Checkout
Low-Stock Alerting	Static Manual Reports	Email Alerts (Paid Tier)	Automated Real-Time low_stock System Alerts
Receipt Printing	Local Printer Drivers	Proprietary POS Printer	Browser-Native Thermal Printing (react-to-print)

Table 3.1 Comparative Analysis Matrix

Major Findings & Research Gaps Identified :

1. **High Licensing Burden:** Small retail stores cannot sustain high SaaS subscriptions.
2. **Unregulated Manager Onboarding:** Legacy systems lack staged approval gates for newly registered staff.
3. **Delayed Stock Auditing:** Batch updates cause out-of-stock incidents during peak billing hours.
4. **Complex User Interfaces:** Legacy software requires extensive staff training.

4. TECHNICAL DESCRIPTION

4.1 Hardware Requirement

This section specifies the empirical technical requirements verified directly from the **MARKETA** codebase (React 19.2.7, Vite 8.1.1, React-to-Print 3.3.0, Recharts 3.10.1, Axios 1.18.1, Inter Google Font, and responsive CSS layout engines) needed to open, run, and operate the web application on a user terminal.

Hardware Component	Minimum System Requirement	Recommended System Requirement
Device Category	Desktop PC, Laptop, POS Terminal, or Tablet	Desktop PC or Dedicated POS Touch Terminal
Processor (CPU)	Dual-Core 1.6 GHz (Intel / AMD / ARM)	Intel Core i3 / AMD Ryzen 3 (Quad-Core 2.0 GHz+)
System Memory (RAM)	2 GB DDR3	4 GB DDR4 or higher
Storage / Disk Space	500 MB free space	2 GB free SSD storage
Display Resolution	1024 × 768 pixels	1920 × 1080 pixels (Full HD)
Barcode Scanner	Optional (Manual search supported)	Handheld USB / Wireless 1D/2D Laser Barcode Scanner
Receipt Printer	Standard PDF Printer	80mm Thermal Receipt Printer (ESC/POS compatible)

Table 4.1 Hardware Requirement

4.2 Software Requirement

Based on codebase analysis of ES2022+ ECMAScript syntax, HTML5 Canvas/SVG, Web Storage APIs, and CSS variable dependencies:

Software Component	Verified Technical Specification
Supported Operating Systems	Windows 10/11, macOS 10.15+, Ubuntu Linux 20.04+, Android 10+ (Tablets), iOS/iPadOS 14+
Recommended Web Browser	Google Chrome
Alternative Supported Browsers	• Microsoft Edge • Mozilla Firefox • Apple Safari
JavaScript Engine	JavaScript (ES6+) Enabled
Web Storage Support	LocalStorage & SessionStorage Enabled
Browser Window Permissions	Pop-up / Print Dialogs Allowed

Table 4.2 Software Requirement

Network & Connectivity Requirements :

Based on codebase analysis of Axios REST API HTTP request payloads and Vite dev/proxy server configurations (vite.config.js port 5173 & Express backend port 5000):

Network Parameter	Verified Technical Specification
Network Type	Local Area Network (LAN), Wi-Fi, or Cellular Data (3G / 4G / 5G)
Network Bandwidth	Minimum 1 Mbps stable connection
Communication Protocol	HTTP / HTTPS (RESTful Architecture)
Default Operating Ports	• Frontend: Port 5173 (Vite Server) / 80 (HTTP) / 443 (HTTPS) • Backend API: Port 5000

Table 4.3 Network & Connectivity Requirements

5. SYSTEM DESIGN AND DEVELOPMENT

5.1 Flow Chart:

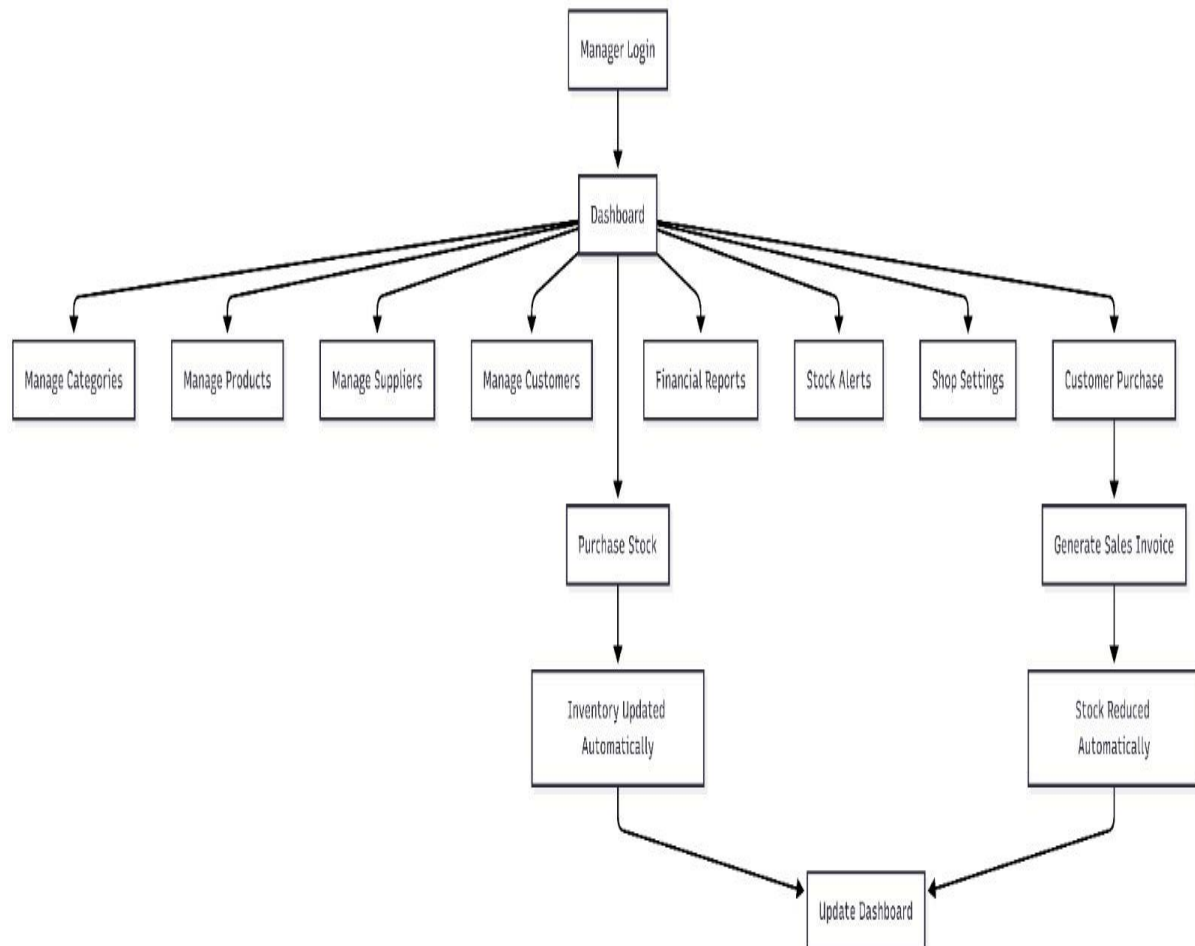


Figure 5.1 Flow Chart

5.2 Data Flow Diagram:

Level 0:-

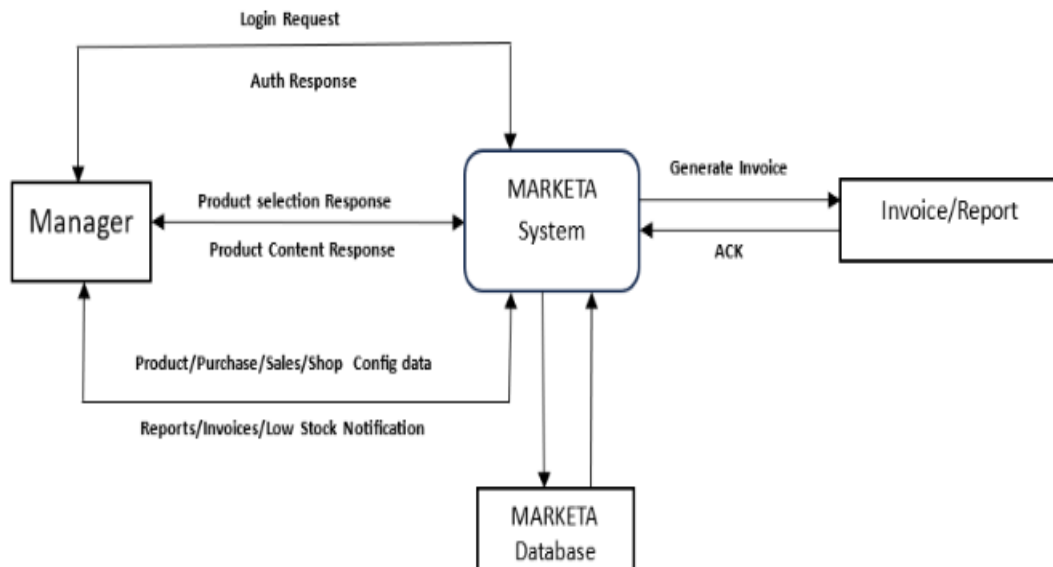


Figure 5.2.1 DFD Level 0

Level 1:-

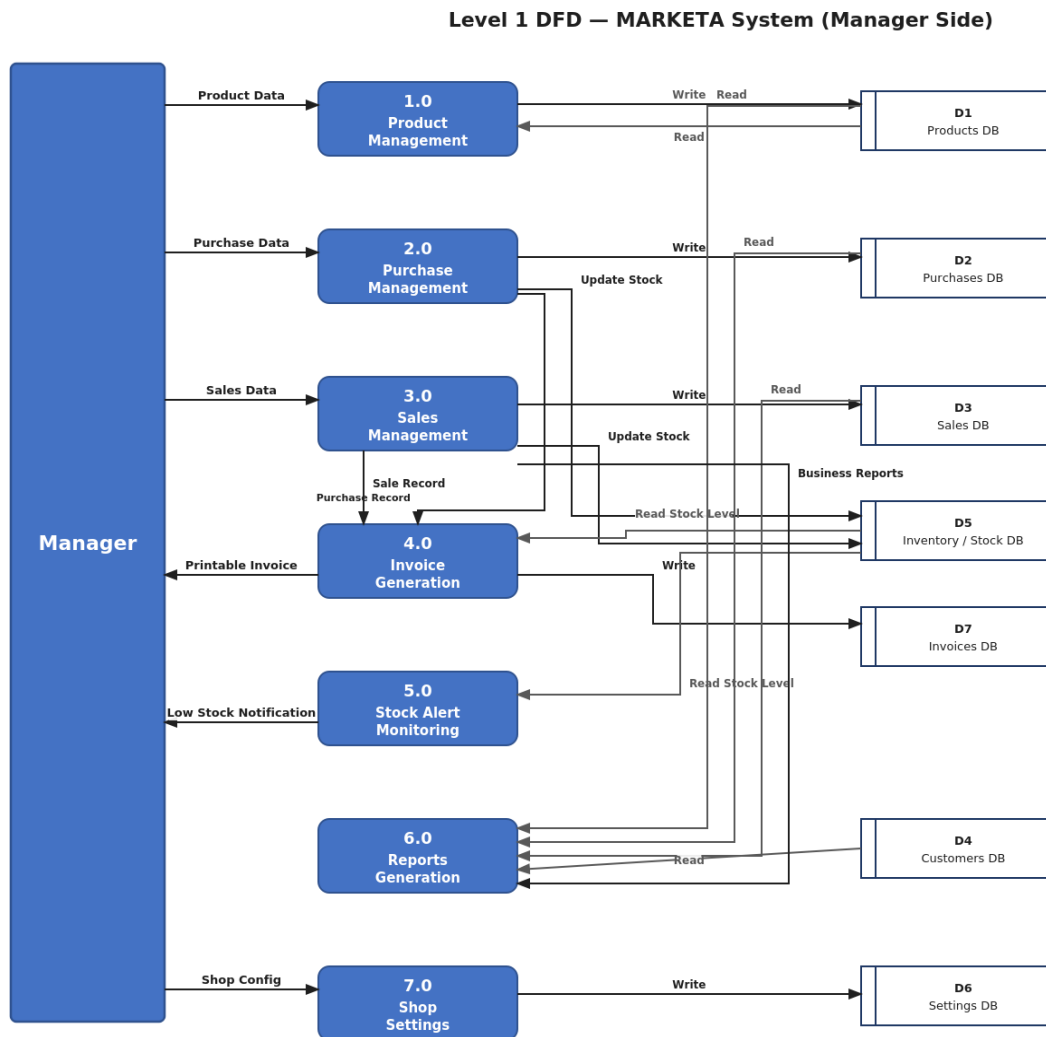


Figure 5.2.2 DFD Level 1

5.3 Use Case Diagram:

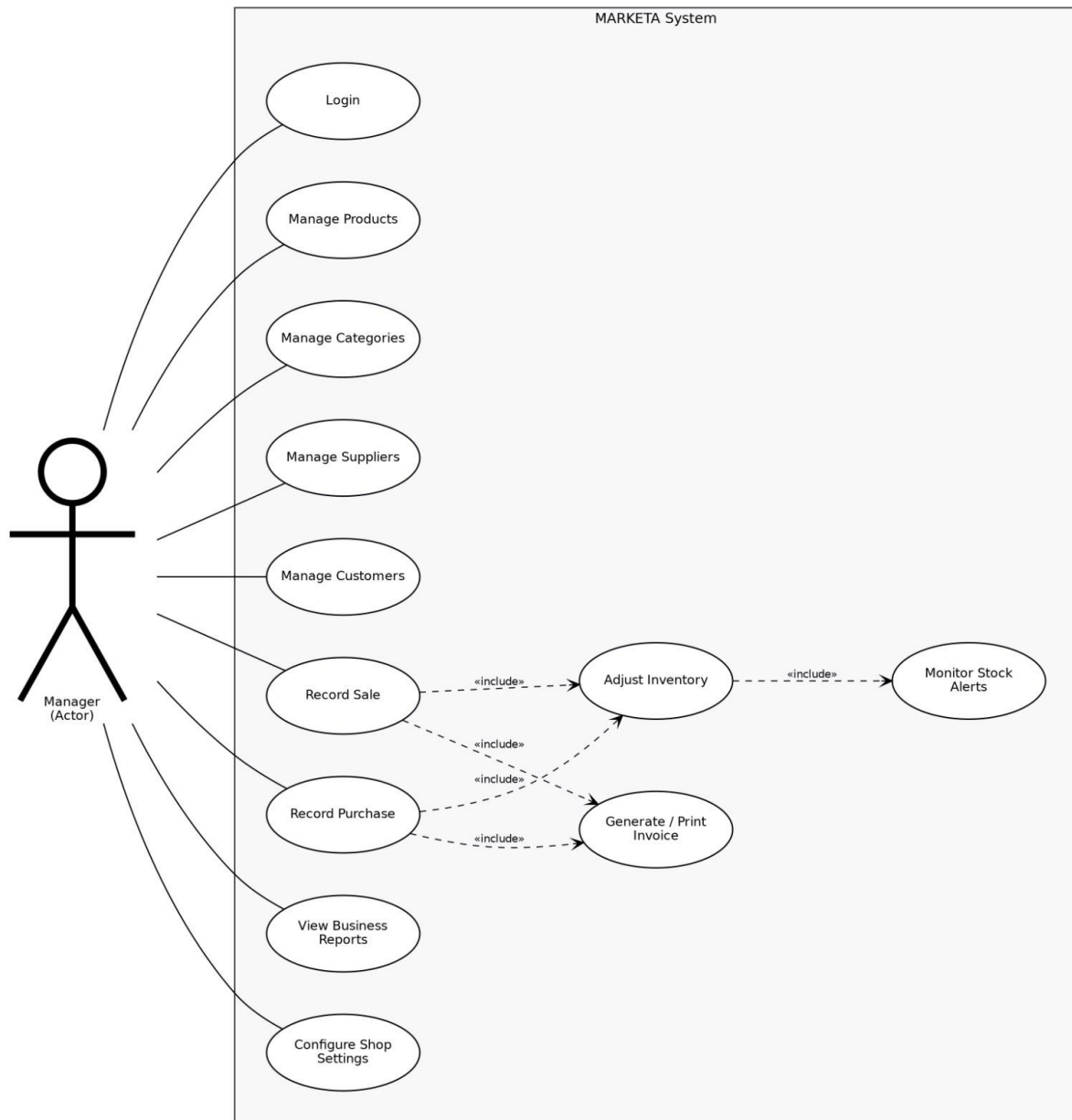


Figure 5.3 Use case diagram

5.4 Sequential Diagram:

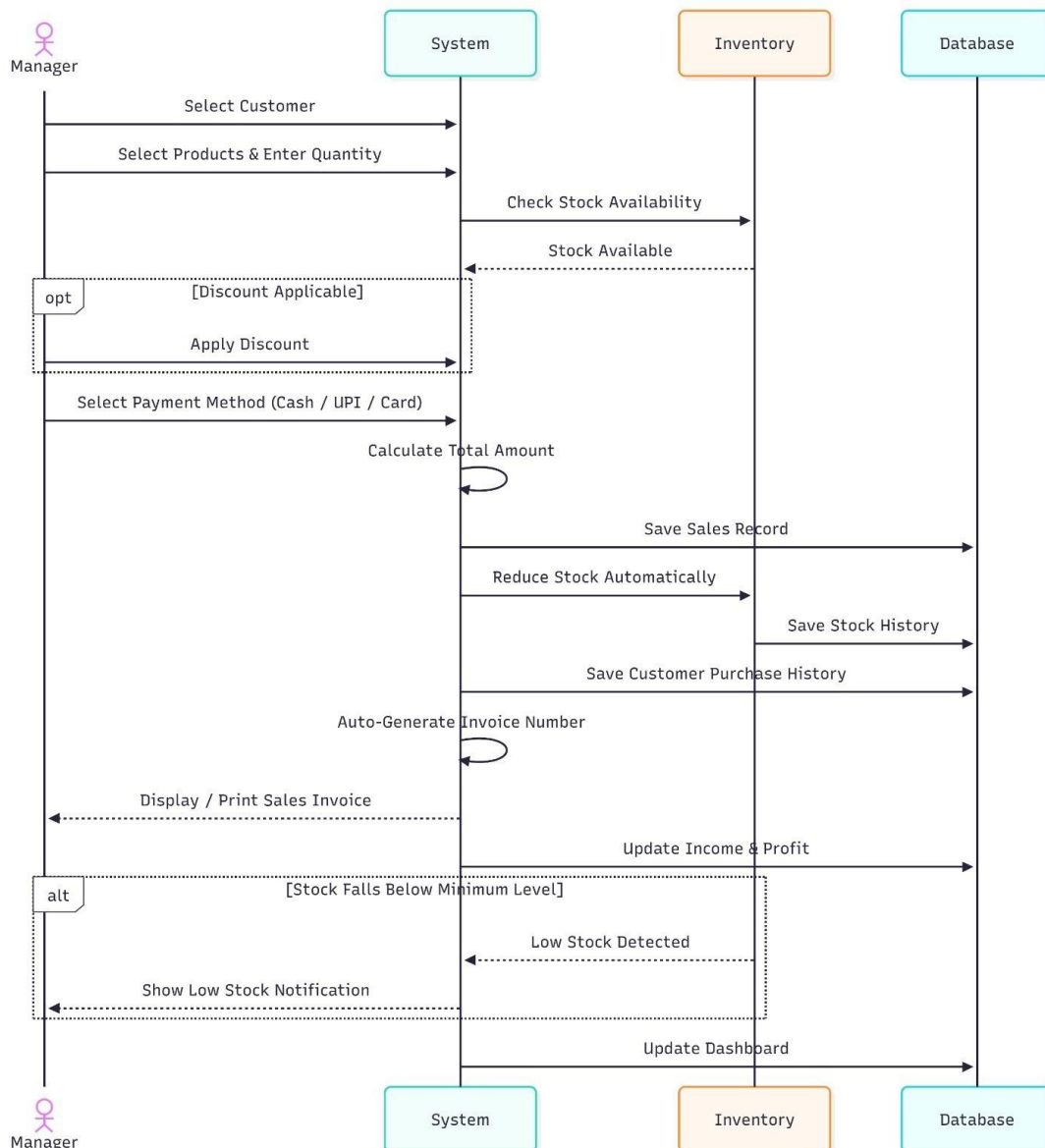


Figure 5.4 Sequential Diagram

5.5 Class Diagram:

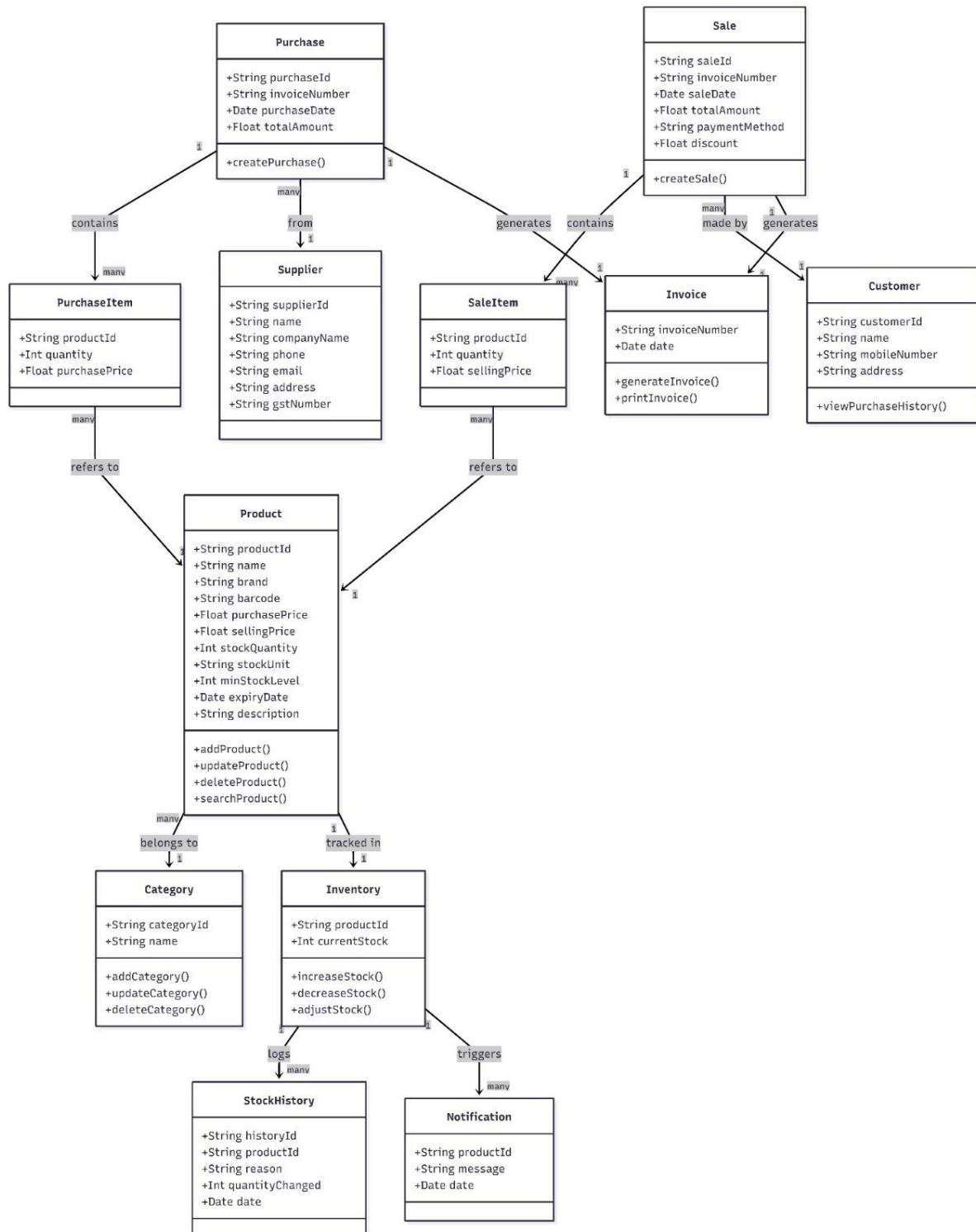


Figure 5.5 Class Diagram

5.6 Activity Diagram:

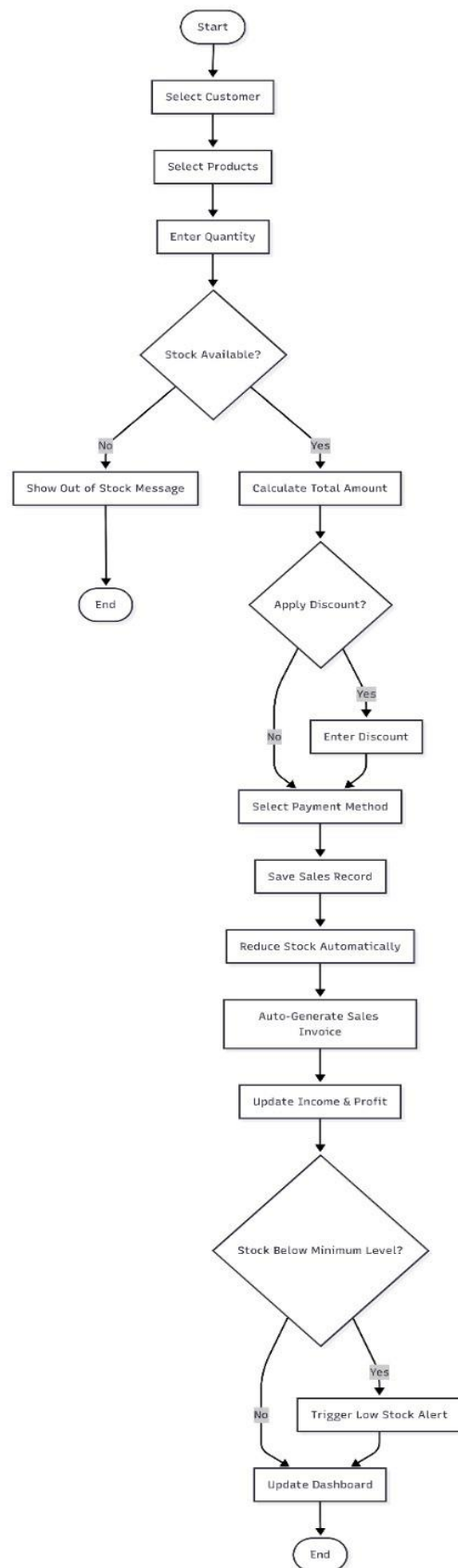


Figure 5.6 Activity Diagram

5.7 Database Design:

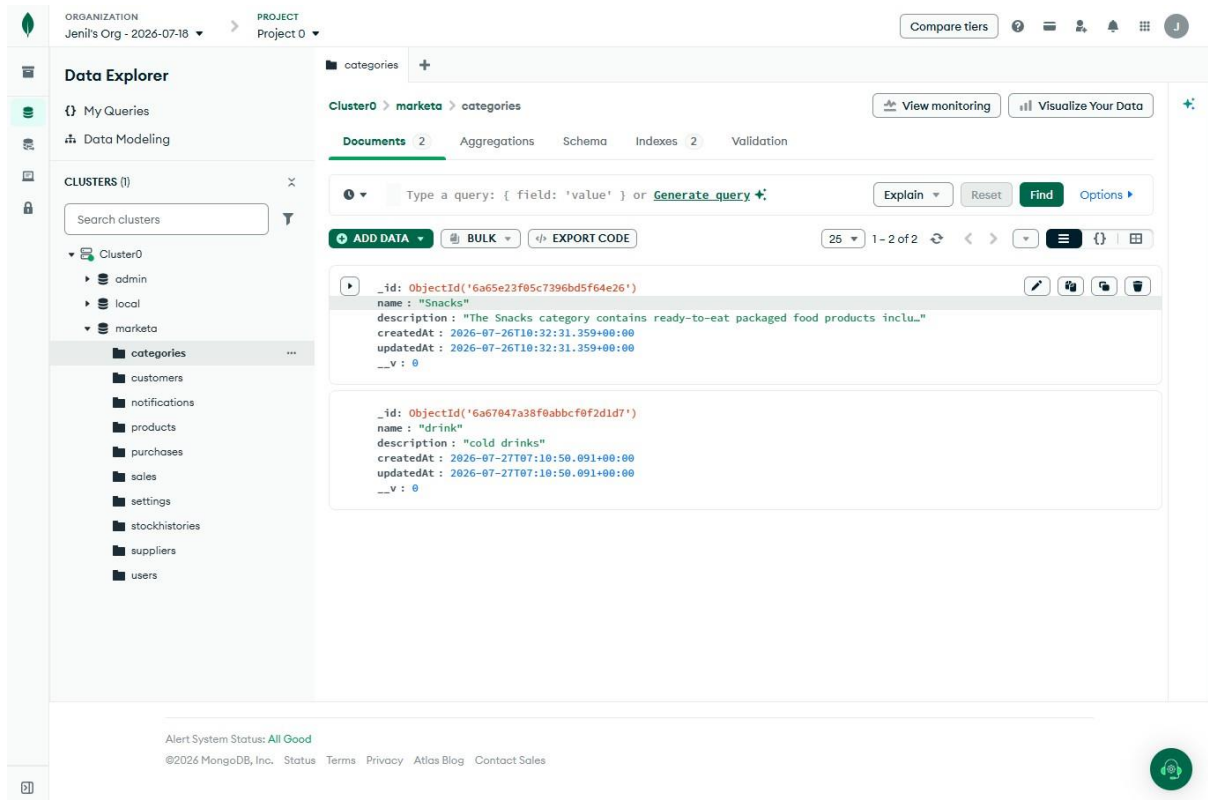


Figure 5.7.1 Categories

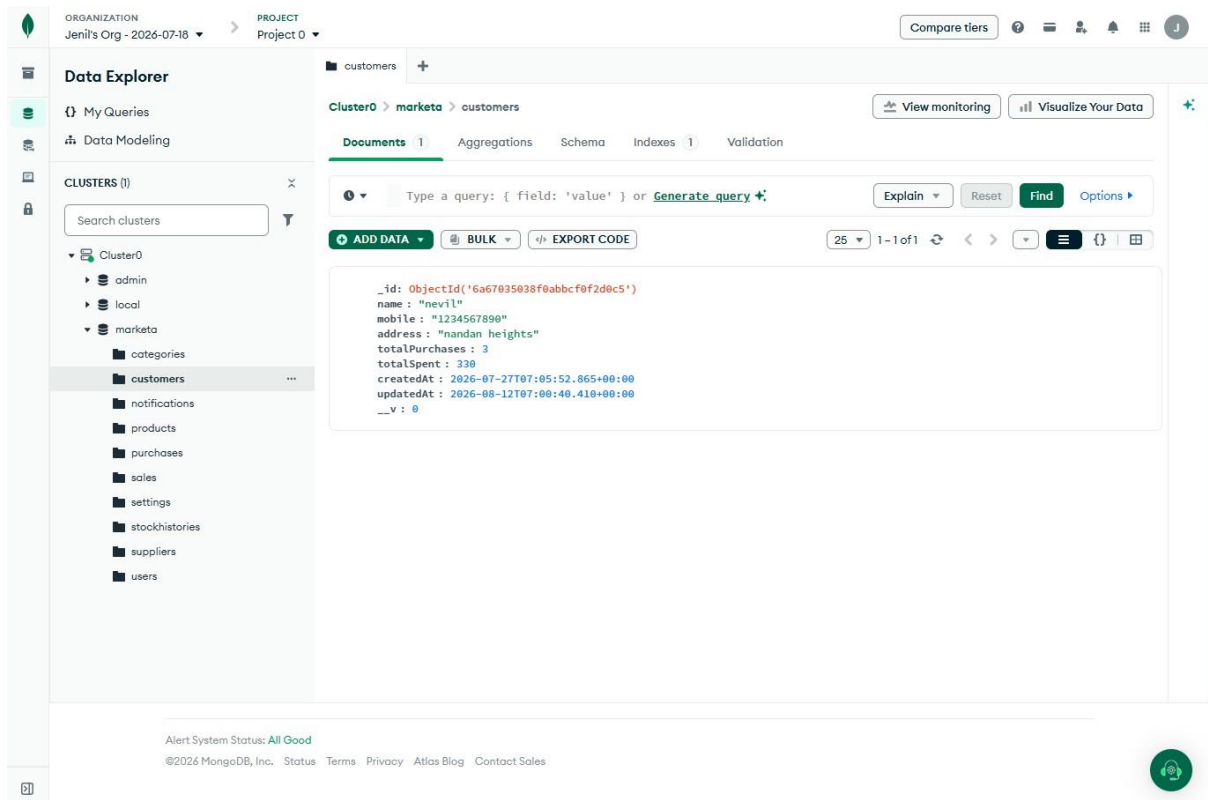


Figure 5.7.2 Customers

Project Title: Marketa

ORGANIZATION: Jenil's Org - 2026-07-18 | PROJECT: Project 0

Compare tiers

Data Explorer

My Queries | Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- local
- marketa
 - categories
 - customers
 - notifications
 - products
 - purchases
 - sales
 - settings
 - stockhistories
 - suppliers
 - users

notifications

Cluster0 > marketa > notifications

Documents 1 | Aggregations | Schema | Indexes 1 | Validation

Type a query: { field: 'value' } or [Generate query](#)

EXPLAIN | RESET | FIND | OPTIONS

ADD DATA | BULK | EXPORT CODE

25 | 1 - 1 of 1

```
{
  "_id": ObjectId('6a7e7ba3e463ae449835d46d'),
  "message": "\ud83d\udc4d Low Stock: thumbsup has 10 left.",
  "type": "low_stock",
  "product": ObjectId('6a6704df38f0abbcf0f2d208'),
  "productName": "thumbsup",
  "isRead": false,
  "createdAt": 2026-08-14T02:21:23.461+00:00,
  "updatedAt": 2026-08-14T02:21:23.461+00:00,
  "__v": 0
}
```

Alert System Status: All Good

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Figure 5.7.3 Notifications

ORGANIZATION: Jenil's Org - 2026-07-18 | PROJECT: Project 0

Compare tiers

Data Explorer

My Queries | Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- local
- marketa
 - categories
 - customers
 - notifications
 - products
 - purchases
 - sales
 - settings
 - stockhistories
 - suppliers
 - users

products

Cluster0 > marketa > products

Documents 5 | Aggregations | Schema | Indexes 1 | Validation

Type a query: { field: 'value' } or [Generate query](#)

EXPLAIN | RESET | FIND | OPTIONS

ADD DATA | BULK | EXPORT CODE

25 | 1 - 5 of 5

```
{
  "_id": ObjectId('6a65e2bd05c7396bd5f64e43'),
  "name": "Gippi",
  "category": ObjectId('6a65e23f05c7396bd5f64e26'),
  "brand": "Balaji",
  "image": "/uploads/image-1785062077381-735115063.jpeg",
  "purchasePrice": 8,
  "sellingPrice": 10,
  "stock": 115,
  "unit": "piece",
  "minStock": 15,
  "description": "Noodles, but not as you know them. They've got the fire, the zing, and..."
  "isLowStock": false,
  "isOutOfStock": false,
  "createdAt": 2026-07-26T10:34:37.393+00:00,
  "updatedAt": 2026-08-04T16:45:04.721+00:00,
  "__v": 0
}
```

```
{
  "_id": ObjectId('6a6704df38f0abbcf0f2d208'),
  "name": "thumbsup",
  "category": ObjectId('6a67047a38f0abbcf0f2d1d7'),
  "brand": "cocacola",
  "purchasePrice": 15,
  "sellingPrice": 20,
  "stock": 10,
  "unit": "piece",
  "minStock": 15,
  "expiryDate": 2026-07-27T00:00:00.000+00:00
}
```

Alert System Status: All Good

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Figure 1.7.4 Products

Project Title: Marketa

ORGANIZATION: Jenil's Org - 2026-07-18 | PROJECT: Project 0

Compare tiers

Data Explorer

My Queries | Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- local
- marketa
 - categories
 - customers
 - notifications
 - products
 - purchases**
 - sales
 - settings
 - stockhistories
 - suppliers
 - users

Cluster0 > marketa > purchases

Documents 6 | Aggregations | Schema | Indexes 2 | Validation

Type a query: { field: 'value' } or [Generate query](#)

EXPLAIN | RESET | FIND | OPTIONS

ADD DATA | BULK | EXPORT CODE

25 | 1 - 6 of 6

```
{
  "_id": "ObjectId('6a65e33f85c7396bd5f64ef6')",
  "invoiceNumber": "PUR-000001",
  "supplier": "ObjectId('6a65e32805c7396bd5f64ede')",
  "items": Array (1),
  "totalAmount": 800,
  "notes": "gipp1",
  "purchaseDate": "2026-07-26T00:00:00.000+00:00",
  "createdAt": "2026-07-26T10:36:47.676+00:00",
  "updatedAt": "2026-07-26T10:36:47.676+00:00",
  "__v": 0
}
```

```
{
  "_id": "ObjectId('6a6704f938f0abbcf0f2d21f')",
  "invoiceNumber": "PUR-000002",
  "supplier": "ObjectId('6a67049c38f0abbcf0f2d1ef')",
  "items": Array (1),
  "totalAmount": 15,
  "notes": "thubsup",
  "purchaseDate": "2026-07-27T00:00:00.000+00:00",
  "createdAt": "2026-07-27T07:12:57.239+00:00",
  "updatedAt": "2026-07-27T07:12:57.239+00:00",
  "__v": 0
}
```

```
{
  "_id": "ObjectId('6a72165e7b7fd6ce3beb03dfd')",
  "invoiceNumber": "PUR-000003",
  "supplier": "ObjectId('6a67049c38f0abbcf0f2d1ef')",
  "items": Array (1),
  "totalAmount": 15,
  "notes": "thubsup",
  "purchaseDate": "2026-07-27T00:00:00.000+00:00",
  "createdAt": "2026-07-27T07:12:57.239+00:00",
  "updatedAt": "2026-07-27T07:12:57.239+00:00",
  "__v": 0
}
```

Alert System Status: All Good

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Figure 5.7.5 Purchases

ORGANIZATION: Jenil's Org - 2026-07-18 | PROJECT: Project 0

Compare tiers

Data Explorer

My Queries | Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- local
- marketa
 - categories
 - customers
 - notifications
 - products
 - purchases
 - sales**
 - settings
 - stockhistories
 - suppliers
 - users

Cluster0 > marketa > sales

Documents 7 | Aggregations | Schema | Indexes 2 | Validation

Type a query: { field: 'value' } or [Generate query](#)

EXPLAIN | RESET | FIND | OPTIONS

ADD DATA | BULK | EXPORT CODE

25 | 1 - 7 of 7

```
{
  "_id": "ObjectId('6a65e36b05c7396bd5f64f48')",
  "invoiceNumber": "INV-000001",
  "customer": null,
  "customerName": "Walk-in Customer",
  "items": Array (1),
  "subtotal": 40,
  "discount": 0,
  "totalAmount": 40,
  "paymentMethod": "cash",
  "saleDate": "2026-07-26T10:37:31.496+00:00",
  "notes": "",
  "createdAt": "2026-07-26T10:37:31.499+00:00",
  "updatedAt": "2026-07-26T10:37:31.499+00:00",
  "__v": 0
}
```

```
{
  "_id": "ObjectId('6a67036e38f0abbcf0f2d0df')",
  "invoiceNumber": "INV-000002",
  "customer": "ObjectId('6a67035038f0abbcf0f2d0c5')",
  "customerName": "nevil",
  "items": Array (1),
  "subtotal": 100,
  "discount": 0,
  "totalAmount": 100,
  "paymentMethod": "cash",
  "saleDate": "2026-07-27T07:06:22.684+00:00",
  "notes": "",
  "createdAt": "2026-07-27T07:06:22.688+00:00",
  "updatedAt": "2026-07-27T07:06:22.688+00:00",
  "__v": 0
}
```

Alert System Status: All Good

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Figure 5.7.6 Sales

Project Title: Marketa

The screenshot displays the MongoDB Atlas Data Explorer interface. On the left, the 'CLUSTERS (1)' sidebar shows a tree view with 'Cluster0' expanded, listing various databases like 'admin', 'local', and 'marketa'. The 'marketa' database is selected, and its 'settings' collection is highlighted. The main panel shows the 'settings' collection with a 'Documents' tab active. It displays a single document with fields such as '_id', 'shopName', 'currency', 'invoiceFooter', 'createdAt', 'updatedAt', 'address', 'gstNumber', 'logo', and 'phone'. The interface includes navigation tabs for 'Documents', 'Aggregations', 'Schema', 'Indexes', and 'Validation'. At the bottom, there is a status bar indicating 'Alert System Status: All Good' and copyright information for MongoDB, Inc.

Figure 5.7.7 Settings

The screenshot displays the MongoDB Atlas Data Explorer interface, focusing on the 'stockhistories' collection. The left sidebar shows the 'CLUSTERS (1)' tree with 'Cluster0' expanded, and 'stockhistories' is selected under the 'marketa' database. The main panel shows the 'stockhistories' collection with a 'Documents' tab active. It displays two documents. The first document represents an 'in' (purchase) transaction, and the second represents an 'out' (sale) transaction. Both documents include fields for '_id', 'product', 'productName', 'type', 'quantity', 'previousStock', 'newStock', 'reason', 'reference', 'notes', 'createdAt', 'updatedAt', and 'v'. The interface includes navigation tabs for 'Documents', 'Aggregations', 'Schema', 'Indexes', and 'Validation'. At the bottom, there is a status bar indicating 'Alert System Status: All Good' and copyright information for MongoDB, Inc.

Figure 5.7.8 Stockhistories

Project Title: Marketa

The screenshot shows the MongoDB Data Explorer interface for the 'marketa' cluster. The left sidebar displays a tree view of clusters and collections. The main area shows the 'suppliers' collection with 3 documents. The document viewer displays the following data:

```
{ "_id": "ObjectId('6a65e32805c7396bd5f64ede')",  
  "name": "ABC",  
  "company": "Balaji",  
  "phone": "1234567890",  
  "email": "abc@gmail.com",  
  "address": "Balaji manufacture",  
  "gstNumber": "GST1254631",  
  "createdAt": "2026-07-26T10:36:24.460+00:00",  
  "updatedAt": "2026-07-26T10:36:24.460+00:00",  
  "__v": 0 },
```

```
{ "_id": "ObjectId('6a67049c38f0abbcf0f2d1ef')",  
  "name": "cocacola",  
  "company": "jkl",  
  "phone": "1234567890",  
  "email": "karannm3656@gmail.com",  
  "address": "nandan heights",  
  "gstNumber": "ASF12",  
  "createdAt": "2026-07-27T07:11:24.916+00:00",  
  "updatedAt": "2026-07-27T07:11:24.916+00:00",  
  "__v": 0 },
```

```
{ "_id": "ObjectId('6a7c18e0ad7352545b816afb')",  
  "name": "gopal",  
  "company": "gopal pvt",  
  "phone": "1234567890",  
  "email": "gopal@gmail.com",  
  "address": "gopal pvt",  
  "gstNumber": "ASF12",  
  "createdAt": "2026-07-27T07:11:24.916+00:00",  
  "updatedAt": "2026-07-27T07:11:24.916+00:00",  
  "__v": 0 },
```

Figure 5.7.9 Suppliers

The screenshot shows the MongoDB Data Explorer interface for the 'marketa' cluster. The left sidebar displays a tree view of clusters and collections. The main area shows the 'users' collection with 3 documents. The document viewer displays the following data:

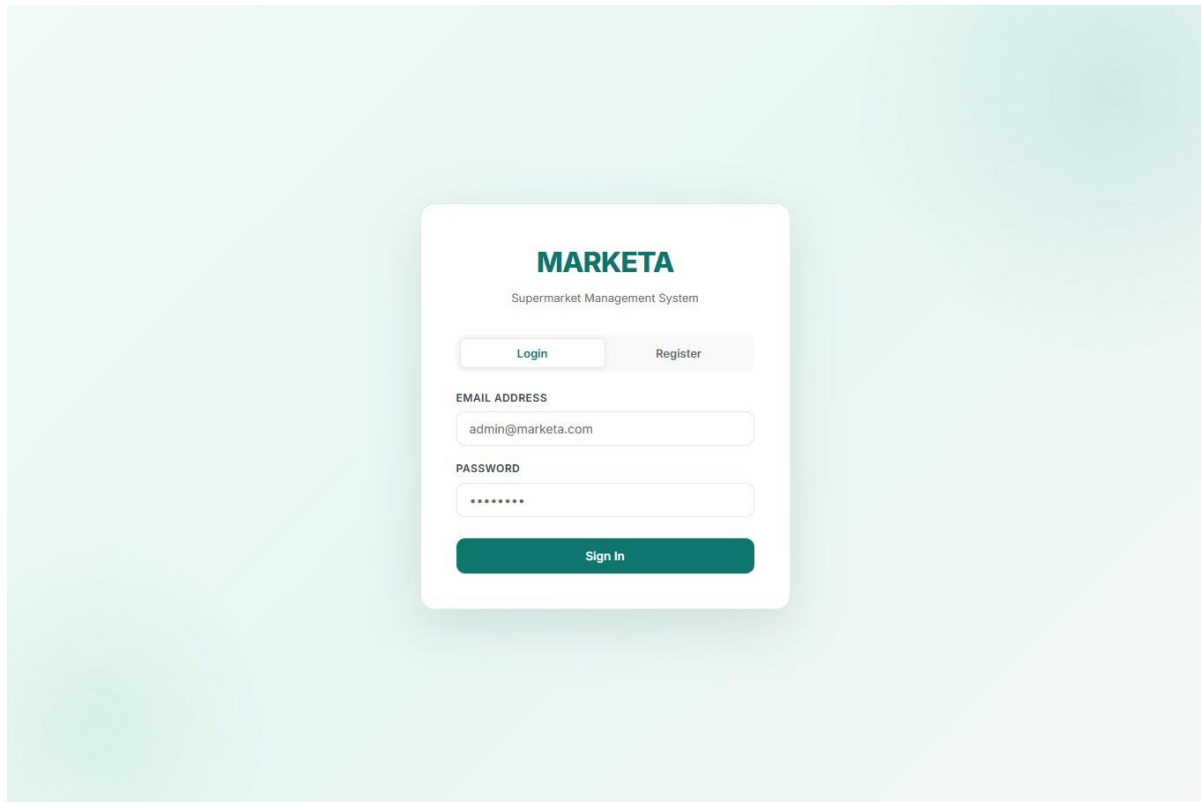
```
{ "_id": "ObjectId('6a7211eb7bfd6ce3beb03b0d')",  
  "name": "jenil",  
  "email": "jenil@gmail.com",  
  "password": "$2a$10$d9v9q0IVyusy5x3xgs0ifV.1D0cQd9aM5KGdW4k5Z5oPKemMatTHLn",  
  "role": "admin",  
  "isApproved": true,  
  "shopName": "123456",  
  "createdAt": "2026-08-04T16:23:07.411+00:00",  
  "updatedAt": "2026-08-04T16:23:07.411+00:00",  
  "__v": 0 },
```

```
{ "_id": "ObjectId('6a7432cf113acd39244f2c31')",  
  "name": "nevil",  
  "email": "nevil@gmail.com",  
  "password": "$2a$10$gK20bhNpx9jphH68PvnXue44AaxwTijx31d282To/G5MA7DiovSt/G",  
  "role": "manager",  
  "isApproved": true,  
  "shopName": "NA",  
  "createdAt": "2026-08-06T07:07:59.723+00:00",  
  "updatedAt": "2026-08-11T04:24:38.908+00:00",  
  "__v": 0 },
```

```
{ "_id": "ObjectId('6a7d515ef498b6e3f17766c5')",  
  "name": "jenil patel",  
  "email": "jenilkothiya.dev@gmail.com",  
  "password": "$2a$10$gK20bhNpx9jphH68PvnXue44AaxwTijx31d282To/G5MA7DiovSt/G",  
  "role": "manager",  
  "isApproved": true,  
  "shopName": "NA",  
  "createdAt": "2026-08-06T07:07:59.723+00:00",  
  "updatedAt": "2026-08-11T04:24:38.908+00:00",  
  "__v": 0 },
```

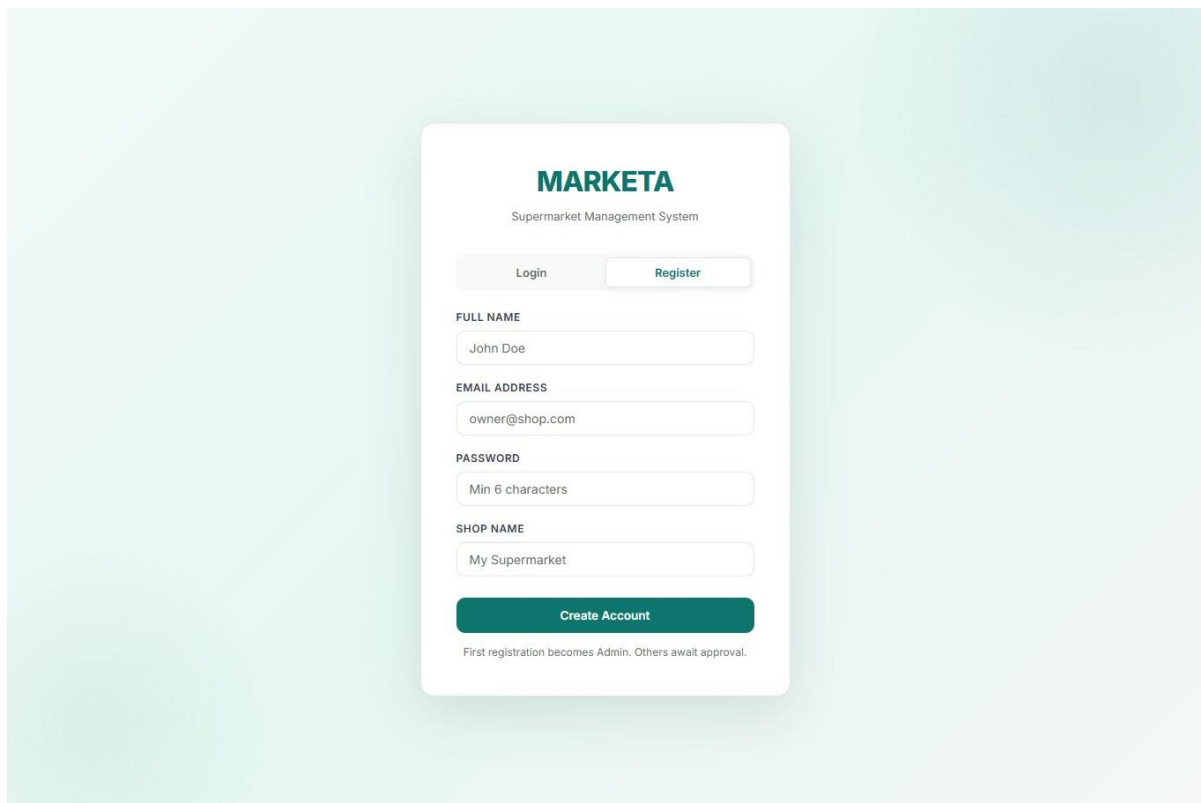
Figure 5.7.10 Users

5.8 Menu-Screen Design:



The login screen for the MARKETA Supermarket Management System. It features a white card centered on a light teal background. The card has the MARKETA logo at the top, followed by the text 'Supermarket Management System'. Below this are two buttons: 'Login' and 'Register'. The 'Login' button is active. Underneath are input fields for 'EMAIL ADDRESS' (containing 'admin@marketa.com') and 'PASSWORD' (containing '*****'). At the bottom is a large green 'Sign In' button.

Figure 5.8.1 Login Screen



The register screen for the MARKETA Supermarket Management System. It features a white card centered on a light teal background. The card has the MARKETA logo at the top, followed by the text 'Supermarket Management System'. Below this are two buttons: 'Login' and 'Register'. The 'Register' button is active. Underneath are input fields for 'FULL NAME' (containing 'John Doe'), 'EMAIL ADDRESS' (containing 'owner@shop.com'), 'PASSWORD' (containing 'Min 6 characters'), and 'SHOP NAME' (containing 'My Supermarket'). At the bottom is a large green 'Create Account' button. Below the button is a small note: 'First registration becomes Admin. Others await approval.'

Figure 5.8.2 Register Screen

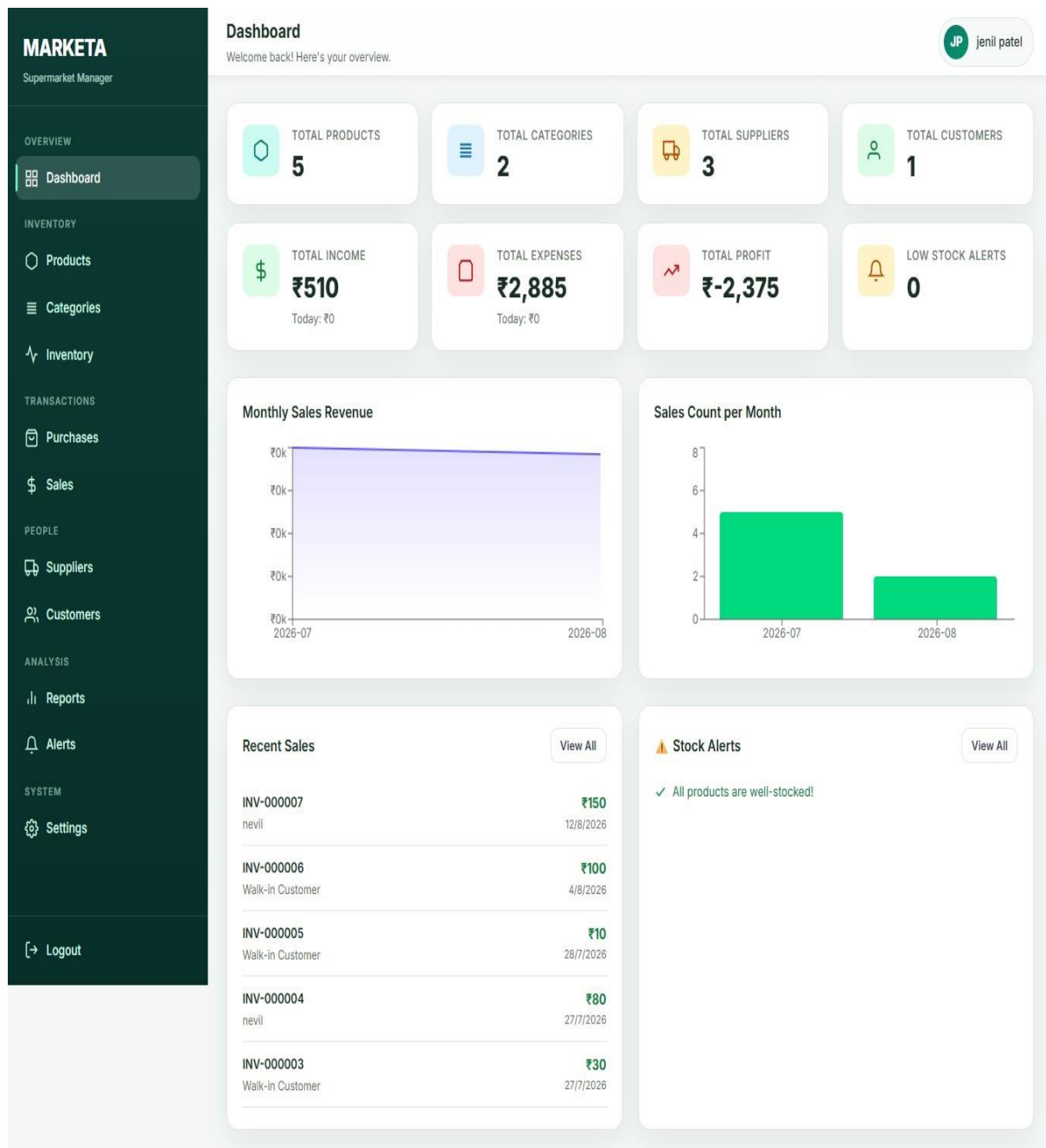


Figure 5.8.3 Dashboard Screen

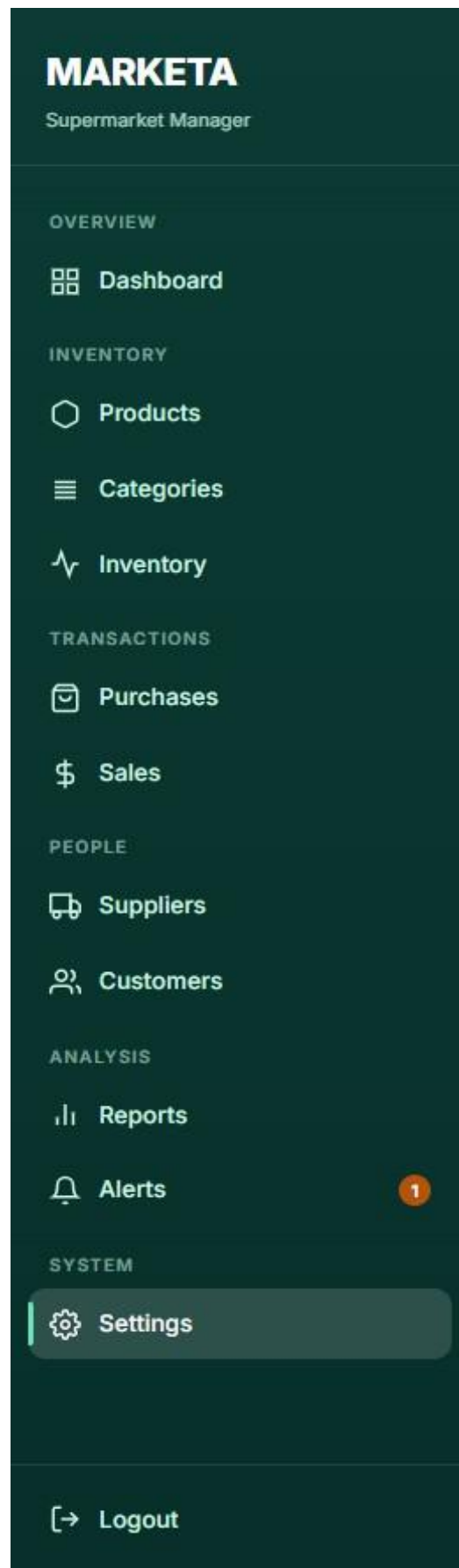


Figure 5.8.4 Modules

Project Title: Marketa

MARKETA
Supermarket Manager

OVERVIEW

Dashboard

INVENTORY

Products

Categories

Inventory

TRANSACTIONS

Purchases

Sales

PEOPLE

Suppliers

Customers

ANALYSIS

Reports

Alerts

SYSTEM

Settings

Logout

Products

Manage your product catalog

JP jenil patel

+ Add Product

Q Search products...

All Categories

All Stock

5 products

IMAGE	PRODUCT	CATEGORY	PURCHASE PRICE	SELLING PRICE	STOCK	STATUS	ACTIONS
	mazaa cocacola	drink	₹13	₹20	150 piece(s)	In Stock	 
	gopal frimes balaji	Snacks	₹5	₹13	10 piece(s)	In Stock	 
	gopal frimes gopal	Snacks	₹8	₹10	135 piece(s)	In Stock	 
	thumbsup cocacola	drink	₹15	₹20	80 piece(s)	In Stock	 
	Gippi Balaji	Snacks	₹8	₹10	115 piece(s)	In Stock	 

Figure 5.8.5 Products Screen

MARKETA
Supermarket Manager

OVERVIEW

Dashboard

INVENTORY

Products

Categories

Inventory

TRANSACTIONS

Purchases

Sales

PEOPLE

Suppliers

Customers

ANALYSIS

Reports

Alerts

SYSTEM

Settings

Logout

Categories

Organize products by category

JP jenil patel

+ Add Category

Q Search categories...

2 categories

#	CATEGORY NAME	DESCRIPTION	CREATED	ACTIONS
1	Snacks	The Snacks category contains ready-to-eat packaged food products including chips, namkeen, popcorn, biscuits, cookies, chocolates, candies, crackers, roasted nuts, and similar snack items available in various flavors, brands, and pack sizes.	26/7/2026	 
2	drink	cold drinks	27/7/2026	 

Figure 5.8.6 Categories

Project Title: Marketa

MARKETA
Supermarket Manager

OVERVIEW

Dashboard

INVENTORY

Products

Categories

Inventory

TRANSACTIONS

Purchases

Sales

PEOPLE

Suppliers

Customers

ANALYSIS

Reports

Alerts

SYSTEM

Settings

Logout

Inventory

Monitor and manage stock levels

JP jenil patel

Stock Adjustment

Current Stock

Stock History

Search products... All Status

PRODUCT	CATEGORY	STOCK	UNIT	MIN LEVEL	STATUS	VALUE
gopal frimes	Snacks	10	piece	5	In Stock	₹130
thumbsup	drink	80	piece	15	In Stock	₹1,600
Gippi	Snacks	115	piece	15	In Stock	₹1,150
gopal frimes	Snacks	135	piece	5	In Stock	₹1,350
mazaa	drink	150	piece	50	In Stock	₹3,000

Figure 5.8.7 Inventory Screen

MARKETA
Supermarket Manager

OVERVIEW

Dashboard

INVENTORY

Products

Categories

Inventory

TRANSACTIONS

Purchases

Sales

PEOPLE

Suppliers

Customers

ANALYSIS

Reports

Alerts

SYSTEM

Settings

Logout

Purchases

Record and track stock purchases

JP jenil patel

+ New Purchase

Record stock purchases from suppliers

INVOICE NO.	SUPPLIER	ITEMS	TOTAL AMOUNT	DATE	ACTIONS
PUR-000006	gopal	1 items	₹400	13/8/2026	View
PUR-000005	ABC	1 items	₹800	4/8/2026	View
PUR-000004	ABC	1 items	₹120	4/8/2026	View
PUR-000003	cocacola	1 items	₹750	4/8/2026	View
PUR-000002	cocacola	1 items	₹15	27/7/2026	View
PUR-000001	ABC	1 items	₹800	26/7/2026	View

Figure 5.8.8 Purchases Screen

Project Title: Marketa

MARKETA
Supermarket Manager

OVERVIEW

- Dashboard

INVENTORY

- Products
- Categories
- Inventory

TRANSACTIONS

- Purchases
- Sales

PEOPLE

- Suppliers
- Customers

ANALYSIS

- Reports
- Alerts

SYSTEM

- Settings

Logout

Sales
Record and track sales

JP jenil patel

+ New Sale

INVOICE NO.	CUSTOMER	ITEMS	DISCOUNT	TOTAL	PAYMENT	DATE	ACTION
INV-000007	nevil	1 items	—	₹150	CASH	12/8/2025	View
INV-000006	Walk-in Customer	1 items	—	₹100	CASH	4/8/2026	View
INV-000005	Walk-in Customer	1 items	—	₹10	CASH	28/7/2026	View
INV-000004	nevil	1 items	—	₹80	UPI	2/1/2026	View
INV-000003	Walk-in Customer	1 items	—	₹30	CASH	27/7/2026	View
INV-000002	nevil	1 items	—	₹100	CASH	27/7/2026	View
INV-000001	Walk-in Customer	1 items	—	₹40	CASH	26/7/2026	View

Figure 5.8.9 Sales Screen

MARKETA
Supermarket Manager

OVERVIEW

- Dashboard

INVENTORY

- Products
- Categories
- Inventory

TRANSACTIONS

- Purchases
- Sales

PEOPLE

- Suppliers
- Customers

ANALYSIS

- Reports
- Alerts

SYSTEM

- Settings

Logout

Suppliers
Manage your supplier contacts

JP jenil patel

+ Add Supplier

Search by name or company...

#	SUPPLIER NAME	COMPANY	PHONE	EMAIL	GST NO.	ACTIONS
1	gopal	gopal pvt	1234567890	gopal@gamil.com	124568	View Edit Delete
2	cocacola	jkl	1234567890	karannm3656@gmail.com	ASF12	View Edit Delete
3	ABC	Balaji	1234567890	abc@gmail.com	GST1254631	View Edit Delete

Figure 5.8.10 Suppliers Screen

MARKETA

Supermarket Manager

OVERVIEW

Dashboard

INVENTORY

Products

Categories

Inventory

TRANSACTIONS

Purchases

Sales

PEOPLE

Suppliers

Customers

ANALYSIS

Reports

Alerts

SYSTEM

Settings

Logout

Customers

Track customer information

JP jenil patel

Customers

Maintain customer purchase records

+ Add Customer

Q Search by name or mobile...

#	CUSTOMER NAME	MOBILE	ADDRESS	TOTAL PURCHASES	TOTAL SPENT	ACTIONS
1	nevil	1234567890	nandan heights	3	₹330	

Figure 5.8.11 Customers Screen

MARKETA

Supermarket Manager

OVERVIEW

Dashboard

INVENTORY

Products

Categories

Inventory

TRANSACTIONS

Purchases

Sales

PEOPLE

Suppliers

Customers

ANALYSIS

Reports

Alerts

SYSTEM

Settings

Logout

Reports

Business insights and analytics

JP jenil patel

Reports

Business insights and analytics

Sales

Purchases

Inventory

Customers

Sales Revenue

Monthly

Top Products

PRODUCT	QTY SOLD	REVENUE
Gippi	18	₹180
thumbsup	9	₹180
gopal frimes	15	₹150

Payment Methods

Figure 5.8.12 Report Screen

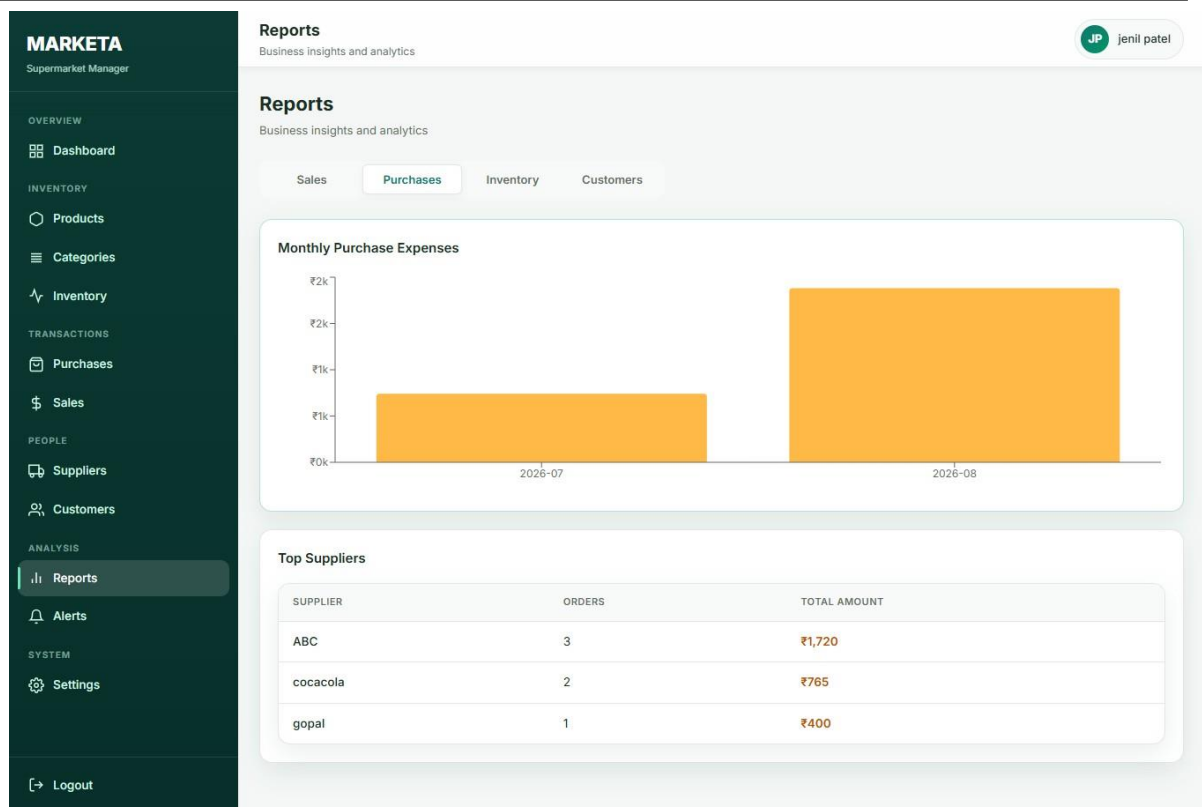


Figure 5.8.13 Report Screen

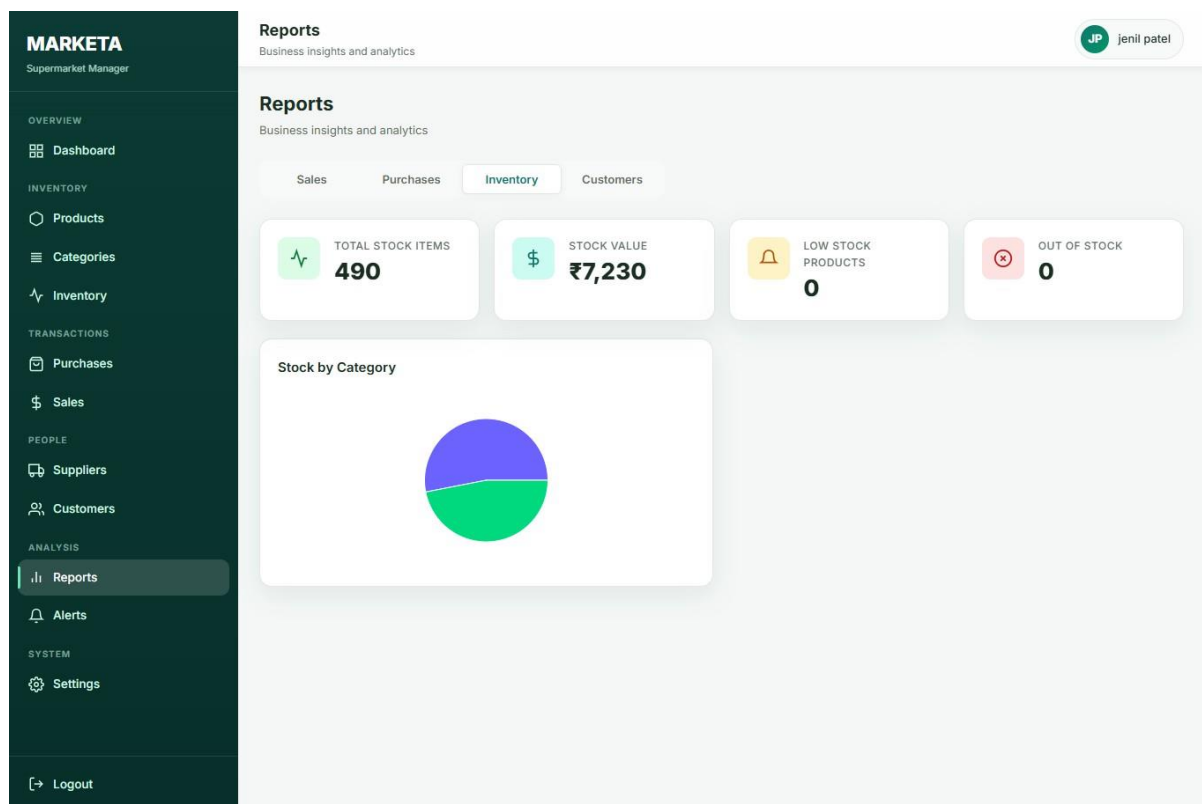


Figure 5.8.14 Report Screen

MARKETA
Supermarket Manager

OVERVIEW

Dashboard

INVENTORY

Products

Categories

Inventory

TRANSACTIONS

Purchases

Sales

PEOPLE

Suppliers

Customers

ANALYSIS

Reports

Alerts

SYSTEM

Settings

Logout

Reports
Business insights and analytics

JP jenil patel

Reports
Business insights and analytics

SalesPurchasesInventoryCustomers

Top Customers by Spending

#	CUSTOMER	MOBILE	TOTAL PURCHASES	TOTAL SPENT
#1	nevil	1234567890	3 orders	₹330

Figure 5.8.15 Reports Screen

MARKETA
Supermarket Manager

OVERVIEW

Dashboard

INVENTORY

Products

Categories

Inventory

TRANSACTIONS

Purchases

Sales

PEOPLE

Suppliers

Customers

ANALYSIS

Reports

Alerts

SYSTEM

Settings

Logout

Notifications
Stock alerts and system messages

JP jenil patel

Notifications
1 unread alert

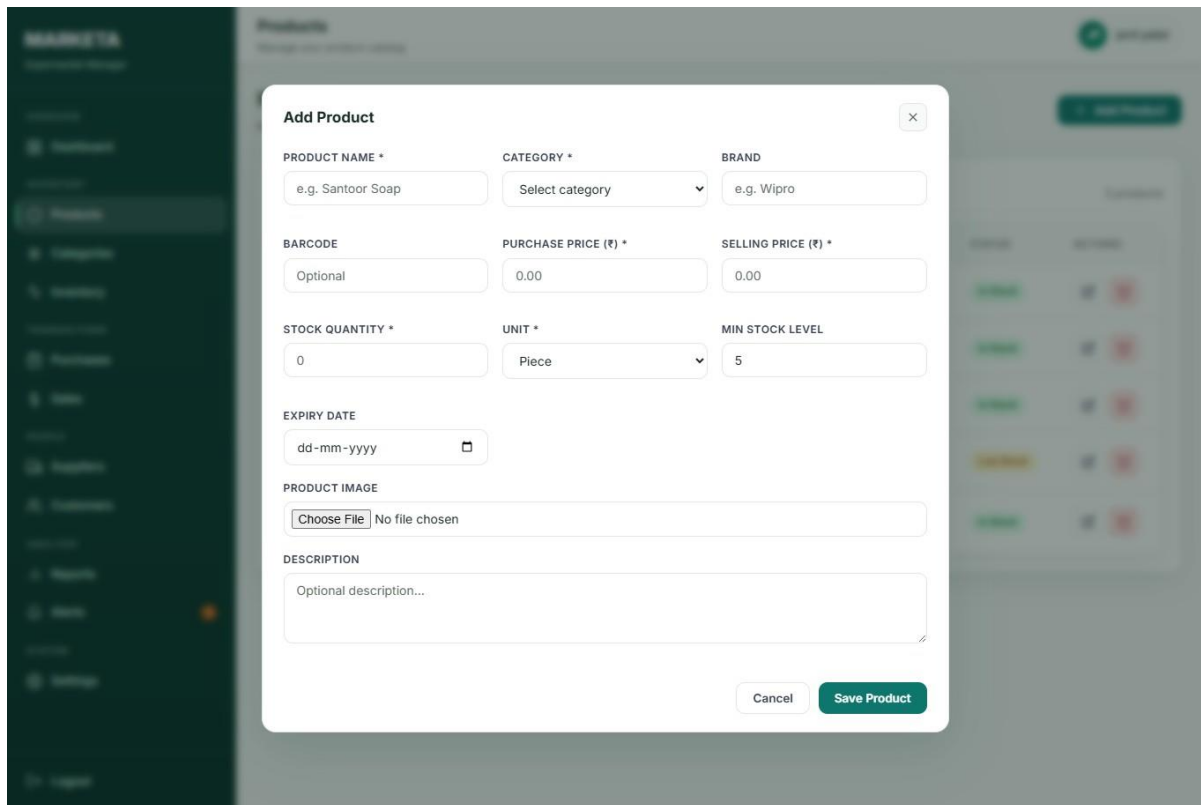
Mark All Read

Low Stock: thumbsup has 10 left.

low stock14/8/2026, 7:51:23 amNEW

✓🗑

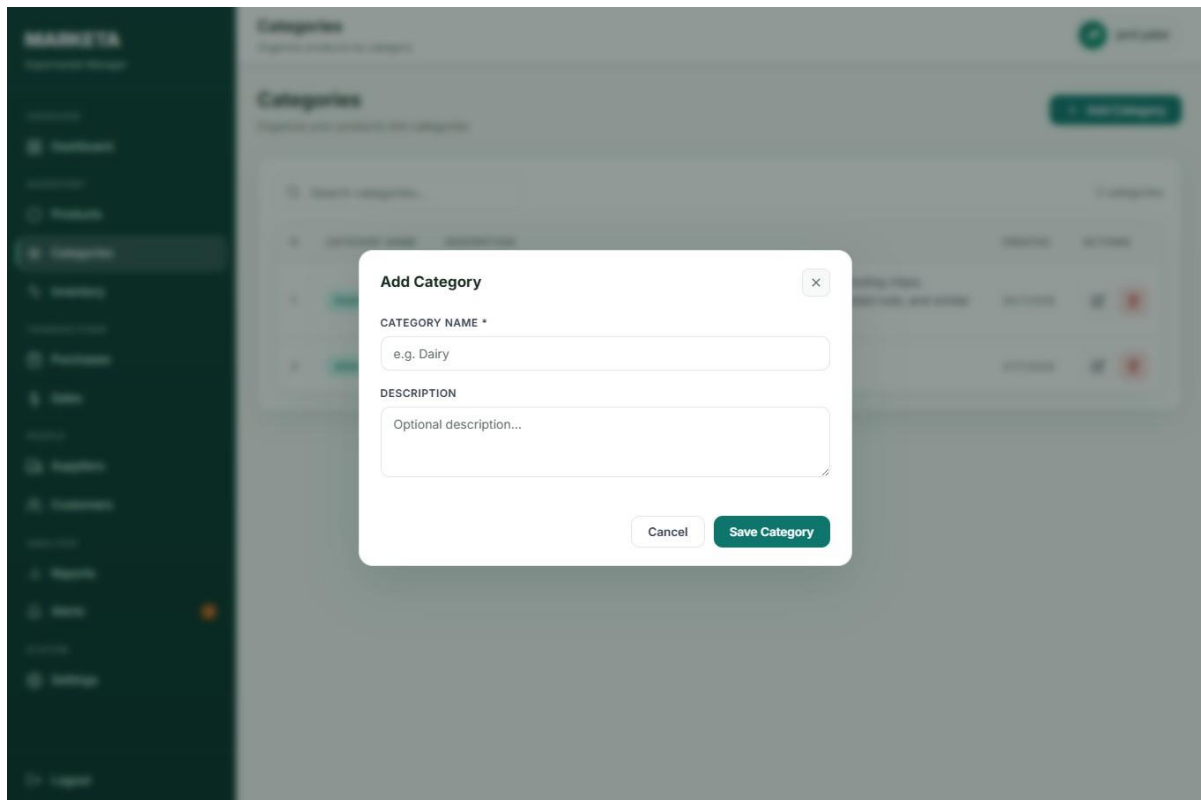
Figure 5.8.16 Alert Screen



The 'Add Product' modal form is displayed over a blurred background of the Marketa application. The form is titled 'Add Product' and includes a close button (X) in the top right corner. It contains several input fields and buttons:

- PRODUCT NAME ***: Text input with placeholder 'e.g. Santoor Soap'.
- CATEGORY ***: Dropdown menu with 'Select category' as the selected option.
- BRAND**: Text input with placeholder 'e.g. Wipro'.
- BARCODE**: Text input with placeholder 'Optional'.
- PURCHASE PRICE (₹) ***: Text input with placeholder '0.00'.
- SELLING PRICE (₹) ***: Text input with placeholder '0.00'.
- STOCK QUANTITY ***: Text input with placeholder '0'.
- UNIT ***: Dropdown menu with 'Piece' as the selected option.
- MIN STOCK LEVEL**: Text input with placeholder '5'.
- EXPIRY DATE**: Text input with placeholder 'dd-mm-yyyy' and a calendar icon.
- PRODUCT IMAGE**: File upload area with a 'Choose File' button and the text 'No file chosen'.
- DESCRIPTION**: Text area with placeholder 'Optional description...'.
- Buttons**: 'Cancel' and 'Save Product' buttons at the bottom right.

Figure 5.8.17 Add Product Screen



The 'Add Category' modal form is displayed over a blurred background of the Marketa application. The form is titled 'Add Category' and includes a close button (X) in the top right corner. It contains several input fields and buttons:

- CATEGORY NAME ***: Text input with placeholder 'e.g. Dairy'.
- DESCRIPTION**: Text area with placeholder 'Optional description...'.
- Buttons**: 'Cancel' and 'Save Category' buttons at the bottom right.

Figure 5.8.18 Add Category Screen

6. SYSTEM TESTING

Software testing was conducted following IEEE 829 Standard Test Documentation guidelines, covering both positive (PASS) and negative (FAIL / Exception handling) scenarios across all modules.

Module 1: User Registration Testing

TestID	Description	Steps	Input Data	Sample Data	Expected Result	Actual Result	Pass/Fail	Comments
TC_REG_001	First user system registration and admin role auto-assignment	1. Open Register page. 2. Enter valid details. 3. Click Register button.	Name, Email, Password, Shop Name	Name: "Admin User" Email: "admin@marketa.com" Password: "Admin@123" ShopName: "Marketa Main"	User created with role: "admin" & isApproved: true. Token returned. HTTP 201 Created.	Account registered as Admin, auto-approved. JWT token generated.	PASS	Positive Test (First user auto-assigned Admin role).
TC_REG_002	Registration attempt with an already registered email	1. Open Register page. 2. Enter existing email. 3. Click Register.	Name, Email, Password, Shop Name	Name: "Duplicate User" Email: "admin@marketa.com" Password: "Pass@123" ShopName: "Branch 2"	HTTP 409 Conflict. Message: "Email already registered". Registration rejected.	Error message displayed, duplicate account creation blocked.	PASS	Negative Test (Email uniqueness validation).
TC_REG_003	Registration submission leaving mandatory Shop Name field blank	1. Open Register page. 2. Fill Name/Email/Password. 3. Leave Shop Name empty. 4. Click Register.	Name, Email, Password, Shop Name	Name: "Manager User" Email: "mgr@marketa.com" Password: "Pass@123" ShopName: ""	HTTP 400 Bad Request error. Registration blocked due to missing shop name.	Account registered without shop name (HTTP 201 Created).	FAIL	Bug #101: Backend schema missing required: true constraint on shopName.

Table 6.1 User Registration Testing

Module 2: User Authentication & Login Testing:

TestID	Description	Steps	Input Data	Sample Data	Expected Result	Actual Result	Pass/Fail	Comments
TC_LOG_001	User login with valid credentials of an approved account	1. Open Login page. 2. Enter valid credentials. 3. Click Login.	Email, Password	Email: "admin@marketa.com" Password: "Admin@123"	HTTP 200 OK. JWT Token generated, user authenticated & redirected to Dashboard.	Login successful, token saved, redirected to /.	PASS	Positive Test (Valid approved user login).
TC_LOG_002	User login attempt with incorrect password	1. Open Login page. 2. Enter valid email. 3. Enter wrong password. 4. Click Login.	Email, Password	Email: "admin@marketa.com" Password: "WrongPass999"	HTTP 401 Unauthorized. Message: "Invalid email or password". Token denied.	Authentication failed with HTTP 401 error message.	PASS	Negative Test (Security credential validation).
TC_LOG_003	Submit login form with completely empty input fields	1. Open Login page. 2. Leave all fields empty. 3. Click Submit.	Email, Password	Email: "" Password: ""	Client-side validation message displayed: "Please fill out all required fields".	Form submitted empty request to API, causing Server 500 error.	FAIL	Bug #102: Frontend missing client-side validation before API dispatch.
TC_LOG_004	Login attempt with pending/unapproved manager account	1. Open Login page. 2. Enter manager credentials. 3. Click Login.	Email, Password	Email: "john_mgr@marketa.com" Password: "Manager@123"	HTTP 403 Forbidden. Message: "Your account is pending admin approval".	Access blocked with HTTP 403 error message.	PASS	Negative Test (Admin approval gate check).

Table 6.2 User Authentication & Login Test Case

Module 3: Product & Inventory Management Testing:

TestID	Description	Steps	Input Data	Sample Data	Expected Result	Actual Result	Pass/Fail	Comments
TC_PRD_001	Add new product with initial stock count and category	1. Log in as Admin. 2. Navigate to Products. 3. Fill product form. 4. Click Save.	Name, Barcode, Category, Price, Stock	Name: "Organic Milk 1L" Barcode: "8901234567890" Category: "<CAT_ID>" Price: 65.00 Stock: 50	HTTP 201 Created. Product saved in DB with stock count 50 & linked Category.	Product added to inventory with stock: 50.	PASS	Positive Test (Product creation & stock setup).
TC_PRD_002	Create product API call without JWT authorization token	1. Send POST to /api/products. 2. Omit Authorization header. 3. Execute request.	Headers, Product Body	Header: None Payload: {"name": "Test Item", "price": 50}	HTTP 401 Unauthorized. Message: "Not authorized to access this route".	Middleware intercepted & rejected request.	PASS	Negative Test (Protected route security check).
TC_PRD_003	Add product with negative price value (price: -50.00)	1. Open Add Product page. 2. Enter negative price. 3. Click Save.	Name, Price, Stock	Name: "Faulty Product" Price: -50.00 Stock: 20	HTTP 400 Bad Request error rejecting negative price inputs.	Product created with negative price -50.00 in database.	FAIL	Bug #103: Product schema lacks min: 0 numerical validation on price.

Table 6.3 Product & Inventory Management Testing

Module 4: Sales & Point of Sale (POS) Testing:

TestID	Description	Steps	Input Data	Sample Data	Expected Result	Actual Result	Pass/Fail	Comments
TC_POS_001	POS sale checkout and automatic stock deduction	1. Select product in POS. 2. Set quantity = 10. 3. Apply discount = 20. 4. Complete checkout.	Product, Quantity, Discount, Payment Method	Product: "Organic Milk 1L" Quantity: 10 Discount: 20.00 PaymentMethod: "cash"	HTTP 201 Created. Invoice generated (INV-20260813-0001), stock deducted from 50 to 40 in DB.	Sale created, invoice generated, stock updated to 40.	PASS	Positive Test (POS Checkout & Auto Stock Deduction).
TC_POS_002	POS sale checkout requesting quantity exceeding available stock	1. Select product (Stock = 40). 2. Set quantity = 100. 3. Click Checkout.	Product, Quantity	Product: "Organic Milk 1L" Quantity: 100	HTTP 400 Bad Request. Message: "Insufficient stock... Available: 40".	Transaction rejected with error message, stock unchanged.	PASS	Negative Test (Inventory stock boundary check).
TC_POS_003	POS checkout with discount greater than sale subtotal	1. Select item (Subtotal = 300). 2. Set discount = 500. 3. Click Checkout.	Items, Discount	Subtotal: 300.00 Discount: 500.00	HTTP 400 Bad Request rejecting discount exceeding total sale amount.	Sale processed with negative total amount (-200.00).	FAIL	Bug #104: POS controller missing discount <= subtotal validation check.
TC_POS_004	Submit POS checkout request with an empty item list	1. Send checkout API call. 2. Pass empty items array. 3. Submit request.	Items Array	Items: []	HTTP 400 Bad Request. Message: "Sale must have at least one item".	Checkout rejected due to empty item array.	PASS	Negative Test (Empty cart validation check).

Table 6.4 Sales & Point of Sales (POS) Test Case

7. CONCLUSION

The **MARKETA Supermarket Management System** successfully addresses the operational bottlenecks experienced by modern retail stores and supermarkets. Developed using the robust **MERN Stack (MongoDB, Express.js, React, Node.js)** following MVC architectural principles, the system delivers an end-to-end management platform.

Key achievements of the system include:

1. **Streamlined Authentication:** Secured multi-role access control with Admin approval workflows for newly registered managers.
2. **Automated Stock Deduction:** POS billing transactions seamlessly calculate taxes, subtotal, discounts, and deduct inventory stock in real time.
3. **Proactive Inventory Monitoring:** Automated low-stock and out-of-stock notifications prevent stockouts and improve supply chain replenishment.
4. **Comprehensive Analytics:** Intuitive data visualizer dashboards provide store management with actionable sales and revenue insights.

By combining zero licensing costs, platform independence, real-time stock sync, and a user-friendly POS interface, **MARKETA** provides an efficient, scalable, and cost-effective digital solution for supermarket management.

8. LEARNING DURING PROJECT WORK

During the development of the RunDSA project, we gained practical knowledge in several areas:

- Learned complete MERN Stack development.
- Improved React.js component-based development.
- Understood backend API development using Node.js and Express.js.
- Learned MongoDB database design and CRUD operations.
- Implemented secure user authentication using JWT.
- Improved knowledge of REST APIs and frontend-backend integration.
- Learned Git and GitHub for version control.
- Enhanced debugging, testing, and problem-solving skills.
- Improved teamwork, communication, and project documentation skills.
- Gained experience in deploying and maintaining full-stack web applications.

8.1 Future Enhancement:

1. **Native WebHID Barcode Scanner Integration:** Implementing browser-level WebHID and WebUSB APIs for zero-latency direct device communication.
2. **Multi-Store Warehouse Synchronization:** Extending schema architecture to support multi-branch supermarket chains with central warehouse tracking.
3. **AI-Driven Demand Forecasting:** Machine Learning integration (ARIMA time-series models) for predicting product demand trends and automated supplier purchase order drafting.
4. **Customer Loyalty & Rewards System:** Points-based customer rewards module integrated into cashier POS checkout.
5. **Offline PWA Support:** Service Worker and IndexedDB integration enabling offline POS billing during network outages.
6. **Integrated Dynamic Payment Gateway:** Instant UPI QR code generation directly on POS billing screen.

9. BIBLIOGRAPHY

9.1 Online References:

1. **Node.js Official Documentation (v18.x LTS)**
URL: <https://nodejs.org/docs/latest-v18.x/api/>
Usage: Referred for ES Modules configuration, asynchronous event handling, and process management.
2. **React 18 Developer Documentation**
URL: <https://react.dev>
Usage: Referred for functional components, custom hooks, Context API, and state management.
3. **Express.js API Reference (v4.x)**
URL: <https://expressjs.com/en/4x/api.html>
Usage: Referred for RESTful routing, custom middleware implementation, and error-handling controllers.
4. **MongoDB & Mongoose Object Data Modeling Guide**
URL: <https://mongoosejs.com/docs/api.html>
Usage: Referred for schema creation, population references, aggregate pipelines, and query optimization.
5. **Vite Frontend Tooling Guide**
URL: <https://vitejs.dev/guide/>
Usage: Referred for frontend bundling, HMR configuration, and asset optimization.
6. **MDN Web Docs (Mozilla Developer Network)**
URL: <https://developer.mozilla.org>
Usage: Referred for REST API standards, HTTP status code specifications, and local storage Web APIs.

9.2 Offline References:

- College lecture notes and practical manuals related to MERN Stack and Data Structures.
- Project guidance provided by the Faculty of Computer Applications, Marwadi University.
- A. Silberschatz, H. F. Korth, and S. Sudarshan, Database System Concepts, 6th ed. New York, NY, USA: McGraw-Hill, 2019.
- R. Elmasri and S. B. Navathe, Fundamentals of Database Systems, 7th ed. Boston, MA, USA: Pearson, 2016.